

Patient Shortage in Histopathology: Directional Shrinkage of Class Centres Gains About One Point on TCGA-UT and Nothing Measurable on BRACS

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Abstract

Correcting the class centres of a five-patient cohort to the centres of the full training pool is worth 10.66 percentage points on TCGA-UT and 8.45 on BRACS, but James-Stein shrinkage toward the cohort's grand mean recovers almost none of that headroom, because it moves each centre along its own discriminative axis only. This report tests the alternative that diagnosis suggests. Directional shrinkage pulls the class centres toward the grand mean along the eigen-directions of the cohort's pooled between-patient covariance, strongly where the spread of the centres is no larger than the noise of a G -patient mean and weakly where it reflects class separation. It uses only the drawn patients, in closed form and with a validation-tuned strength. On TCGA-UT the tuned arm gains 1.29 points at five patients (95% interval: 0.99 to 1.57), meeting the prespecified criterion of at least one point with an interval above zero, and recovers 12 percent of the headroom; the gain falls to 0.41 points at twenty patients. On BRACS it gains 0.36 points (-0.31 to 1.11), which is inconclusive, and it trails radial shrinkage. A geometric diagnostic on the fitted cohorts shows that the gain does not come from better centres: the closed-form arm removes only 4 percent of the squared centre error, the strengths that validation selects on TCGA-UT increase it by 33 to 62 percent, and per-draw gains are uncorrelated with error reduction. The remedy behaves instead like a first-moment form of cohort whitening, which it trails by about one point on TCGA-UT and matches at zero on BRACS. The centre-error headroom remains essentially unrecovered from the cohort alone.

1 Motivation

Class-centre error carries most of the accuracy benefit of additional training patients. Shifting the class centres of a five-patient cohort to the centres of the full training pool gains 10.66 percentage points on TCGA-UT and 8.45 on BRACS (Queisler, 2026b,a). That shift needs the pool, so it measures headroom rather than offering a remedy. The obvious remedy that uses only the drawn patients, James-Stein shrinkage of every class centre toward the cohort's grand mean, recovers -0.028 of that headroom on TCGA-UT and 0.131 on BRACS (Queisler, 2026c).

The diagnosis of that failure points to a specific alternative. Radial shrinkage moves each class centre along a single direction, the ray from the grand mean to the centre. Only a third of the squared centre error lies along that ray, and the ray is also the class's own discriminative axis, so the estimator removes true separation together with error (Queisler, 2026c). A better estimator would act on many directions at once and would treat them differently: it would shrink strongly where the spread of the class centres is no larger than the noise a G -patient mean carries, and weakly where the spread is mostly real class separation. The cohort's own patients supply both quantities, because the pooled within-class covariance of the patient means estimates the noise along every direction it spans.

This report asks one question: *does shrinking class centres along the cohort's own between-patient directions, with a per-direction strength set by a signal-to-noise estimate, recover part of*

the pool-centre headroom that radial shrinkage misses? The answer is judged at five patients per class, separately on each dataset, against a practical threshold of one percentage point that was fixed before fitting.

2 Idea

Write the patch features of class c as $\mathbf{x}_{ik} = \boldsymbol{\mu}_c + \mathbf{u}_i + \mathbf{e}_{ik}$, with class centre $\boldsymbol{\mu}_c$, patient offset $\mathbf{u}_i$, and within-patient residual $\mathbf{e}_{ik}$ (Queisler, 2026b). A cohort of G patients per class estimates $\boldsymbol{\mu}_c$ by the mean $\hat{\boldsymbol{\mu}}_c$ of its G patient means $\mathbf{m}_{ci}$. Pooling the deviations $\mathbf{m}_{ci} - \hat{\boldsymbol{\mu}}_c$ over all C classes at $C(G - 1)$ degrees of freedom gives a between-patient covariance with orthonormal eigenvectors $\mathbf{u}_j$ and eigenvalues λ_j , truncated at a relative tolerance of 10^{-9} exactly as in the cohort-basis whitening arm of Queisler (2026d). Its rank is at most $C(G - 1)$.

Let $\bar{\boldsymbol{\mu}} = C^{-1} \sum_c \hat{\boldsymbol{\mu}}_c$ be the cohort’s grand mean. Along eigen-direction j , the centred class centres have coordinates $z_{cj} = \mathbf{u}_j^\top (\hat{\boldsymbol{\mu}}_c - \bar{\boldsymbol{\mu}})$. Their spread across classes mixes true separation with the noise of a G -patient mean, whose variance along $\mathbf{u}_j$ is $n_j = \lambda_j/G$. A method-of-moments estimate of the separation is

$$s_j = \max\left(0, \frac{1}{C-1} \sum_{c=1}^C z_{cj}^2 - n_j\right), \quad a_j(\alpha) = \min\left(1, \alpha \frac{n_j}{s_j + n_j}\right), \quad (1)$$

with $a_j = 0$ when $s_j + n_j = 0$. The shrunk centres are

$$\tilde{\boldsymbol{\mu}}_c(\alpha) = \hat{\boldsymbol{\mu}}_c - \sum_j a_j(\alpha) z_{cj} \mathbf{u}_j. \quad (2)$$

At $\alpha = 1$ each coordinate is pulled toward the grand mean by the empirical-Bayes fraction $n_j/(s_j + n_j)$: completely where the cohort sees no separation beyond noise, barely where separation dominates. Because $\sum_c z_{cj} = 0$, the grand mean is preserved, and components of $\hat{\boldsymbol{\mu}}_c - \bar{\boldsymbol{\mu}}$ outside the span of the eigenbasis are left untouched. Every ingredient is computed from the drawn patients alone.

Equation 2 differs from radial shrinkage in two ways that the diagnosis of Queisler (2026c) asks for. It uses one weight per direction rather than one per class, so it can suppress error along noisy directions while leaving discriminative ones intact. And it uses a covariance pooled across classes, so its directions are not confined to the $C - 1$ dimensional span of the centred class centres. Neither property guarantees lower risk: the pooled covariance assumes all classes share their patient-level noise, the eigenbasis is itself estimated from $C(G - 1)$ deviations in 2,560 dimensions, and the separation estimate s_j uses only $C - 1$ degrees of freedom per direction.

3 Design

The study keeps the setting of Queisler (2026b) on TCGA-UT and of Queisler (2026a) on BRACS. TCGA-UT contributes 30 cancer types and BRACS its seven lesion classes, both with frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), three fixed patient-disjoint splits, and multinomial logistic regression whose regularization strength is selected on validation patient-macro balanced accuracy. For each split and each of ten draws, twenty eligible patients per class are sampled, and nested cohorts of five patients with 32 patches, ten with sixteen, and twenty with eight are formed, so every classifier trains on 160 patches per class at a single patient count for all classes.

Two new arm families enter at each $G \in \{5, 10, 20\}$ (Table 1). Both apply Equation 2 to the training features only, by the centre shift of the pool-centre arm: one vector per class is added to every training patch, so the class mean reaches its target while each patch keeps its offset

Arm	Source	Training features
<i>RG</i>	reused	Real cohort of $G \in \{5, 10, 20\}$ patients
<i>CG</i>	reused	Real cohort with class centres shifted to the pool centres
<i>SG, StG</i>	reused	Radial shrinkage toward the grand mean, closed form and tuned (Queisler, 2026c)
<i>RWcG</i>	reused	Real cohort whitened along the cohort’s own between-patient basis (Queisler, 2026d,a)
<i>AG</i>	new	Directional shrinkage at $\alpha = 1$ (Equation 2)
<i>AtG</i>	new	<i>AG</i> with α selected on validation from $\{0, 0.5, 1, 2, 4\}$

Table 1: The two new families use only the drawn patients. *RG* and *CG* bracket the available headroom; *SG*, *StG*, and *RWcG* are the cohort-only remedies already tested. *AtG* includes $\alpha = 0$, so its grid contains the unshrunk cohort.

from the class mean. Validation and test features are untouched. The reused arms are the exact fits of the source studies on the same cohorts, so every contrast is paired draw by draw.

The tuned family selects α jointly with the regularization strength on validation patient-macro balanced accuracy, with ties resolved toward the larger α . The $\alpha = 0$ candidate is the *RG* fit itself and the $\alpha = 1$ candidate is the *AG* fit, so only the three remaining strengths are fit anew. The study adds 180 classifiers per dataset, 360 in total. Test accuracy never entered selection, and the grid was not extended after test results were inspected.

4 Evaluation

The endpoint is patient-macro balanced accuracy on the test partition, pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates, so every contrast is paired, and intervals are 95 percent percentile intervals.

The primary contrast is *At5*–*R5*, read separately on each dataset. The remedy counts as a practical success on a dataset if the estimated gain is at least one percentage point and its paired interval lies entirely above zero. A claim across both cohorts requires success on both. Secondary quantities are the gain of the fixed arm, gains at ten and twenty patients, the contrasts against radial shrinkage and cohort whitening, the *recovery* $\rho_G = (\text{AtG} - \text{RG})/(\text{CG} - \text{RG})$ of the pool-centre headroom, and the *shrinkage share* $1 - \Delta^{\text{At}}/\Delta^{\text{R}}$ of the patient-count gap $\Delta^X = XG' - XG$. Recovery and shares are computed per replicate.

After fitting, the pool centres serve a diagnostic that explains rather than tunes: on the exact fitted cohorts, it measures how much of the squared centre error $\sum_c \|\hat{\mu}_c - \mu_c^{\text{pool}}\|^2$ each estimator removes, where the removal falls relative to the discriminant subspace spanned by the centred pool centres, and how much of the correction leaves the span of the cohort’s own centred class centres.

5 Results

Directional shrinkage meets the prespecified success criterion on TCGA-UT and fails it on BRACS. On TCGA-UT, the tuned arm gains 1.29 points at five patients (95% interval: 0.99 to 1.57), the fixed arm 1.03 points (0.83 to 1.22), and both gains shrink with the patient count, to 0.41 and 0.21 points at twenty (Table 2). On BRACS, the tuned arm gains 0.36 points at five patients (−0.31 to 1.11), the fixed arm 0.69 (−0.33 to 1.76), and at twenty patients both are

slightly negative. The primary contrast therefore succeeds on one dataset only, and no claim across both cohorts is supported.

Arm family	$G = 5$	$G = 10$	$G = 20$
<i>TCGA-UT</i>			
Real (R)	62.48 [61.24, 63.65]	67.79 [66.47, 69.00]	71.89 [70.55, 73.06]
Directional, closed form (A)	63.51 [62.24, 64.67]	68.27 [66.95, 69.45]	72.10 [70.76, 73.28]
Directional, tuned (At)	63.77 [62.51, 64.90]	68.50 [67.20, 69.67]	72.30 [70.98, 73.45]
Pool centres (C)	73.14 [71.62, 74.50]	74.47 [73.01, 75.78]	75.55 [74.11, 76.85]
Gain A – R	1.03 [0.83, 1.22]	0.48 [0.39, 0.57]	0.21 [0.16, 0.25]
Gain At – R	1.29 [0.99, 1.57]	0.71 [0.52, 0.90]	0.41 [0.28, 0.54]
Headroom C – R	10.66 [9.78, 11.47]	6.68 [5.94, 7.42]	3.66 [3.06, 4.30]
<i>BRACS</i>			
Real (R)	36.18 [33.46, 38.80]	38.49 [35.87, 41.22]	40.97 [38.50, 43.62]
Directional, closed form (A)	36.87 [34.45, 39.27]	38.86 [36.34, 41.47]	40.63 [37.97, 43.50]
Directional, tuned (At)	36.54 [34.02, 39.11]	38.84 [36.32, 41.47]	40.85 [38.18, 43.62]
Pool centres (C)	44.63 [41.33, 48.05]	45.11 [41.85, 48.51]	45.50 [42.20, 48.76]
Gain A – R	0.69 [−0.33, 1.76]	0.37 [−0.46, 1.17]	−0.35 [−1.34, 0.55]
Gain At – R	0.36 [−0.31, 1.11]	0.34 [−0.29, 1.01]	−0.12 [−0.92, 0.63]
Headroom C – R	8.45 [6.31, 10.84]	6.61 [4.85, 8.58]	4.53 [2.77, 6.37]

Table 2: Directional shrinkage gains about one point at five TCGA-UT patients and nothing distinguishable from zero on BRACS. Accuracy entries are test patient-macro balanced accuracy in percent; gains and headroom are paired contrasts in percentage points; bold entries are the primary contrast. Intervals are 95% bootstrap intervals over test patients and draws. The R and C rows are those of Queisler (2026b) and Queisler (2026a).

The TCGA-UT success is narrow in two respects. Its point estimate clears the one-point threshold, but the lower interval bound of 0.99 does not, so the data exclude a zero gain but not a gain slightly below the threshold. And the gain is small against the headroom: at five patients the tuned arm captures 12 percent of what correct centres are worth.

5.1 Headroom recovery and the patient-count gap

On TCGA-UT, the tuned arm recovers 0.121 (0.093 to 0.148) of the pool-centre headroom at five patients and 0.106 (0.078 to 0.135) at ten; the fixed arm recovers 0.097 and 0.071 (Table 3). These are the first cohort-only centre corrections in this series with recovery intervals clearly above zero on TCGA-UT, where radial shrinkage recovered −0.028 and 0.024. The gain decays with the patient count roughly as the headroom does, so the five-to-twenty gap narrows from 9.41 to 8.53 points, a shrinkage share of 0.09 (0.06 to 0.12). On BRACS, recovery is 0.043 (−0.041 to 0.124) at five patients and 0.052 at ten, and the gap share of 0.10 (−0.12 to 0.29) is indistinguishable from zero.

5.2 Comparison with the other cohort-only remedies

Against radial shrinkage, directional shrinkage is better on TCGA-UT at every patient count and worse on BRACS (Figure 1). On TCGA-UT, At exceeds tuned radial shrinkage by 1.14 points at five patients (0.84 to 1.43), 0.57 at ten, and 0.27 at twenty, all with intervals above zero. On BRACS, it falls behind tuned radial shrinkage by 0.62 points at five patients (−1.47 to 0.15) and 0.61 at twenty, and behind the closed-form radial arm by 0.87 points at twenty (−1.70 to −0.03). The two datasets thus reverse the ranking of the two first-moment estimators.

	TCGA-UT	BRACS
Recovery ρ_5, A	0.097 [0.077, 0.116]	0.082 [−0.045, 0.191]
Recovery ρ_5, At	0.121 [0.093, 0.148]	0.043 [−0.041, 0.124]
Recovery ρ_{10}, A	0.071 [0.059, 0.086]	0.056 [−0.071, 0.180]
Recovery ρ_{10}, At	0.106 [0.078, 0.135]	0.052 [−0.049, 0.153]
Gap, real $\Delta_{5 \rightarrow 20}^R$	9.41 [8.71, 10.12]	4.80 [2.88, 6.75]
Gap, tuned $\Delta_{5 \rightarrow 20}^{At}$	8.53 [7.82, 9.24]	4.31 [2.48, 6.07]
Share, A, $5 \rightarrow 20$	0.09 [0.07, 0.11]	0.22 [−0.03, 0.46]
Share, At, $5 \rightarrow 10$	0.11 [0.05, 0.16]	0.01 [−0.48, 0.35]
Share, At, $10 \rightarrow 20$	0.07 [0.03, 0.11]	0.19 [−0.24, 0.54]
Share, At, $5 \rightarrow 20$	0.09 [0.06, 0.12]	0.10 [−0.12, 0.29]

Table 3: On TCGA-UT directional shrinkage captures a tenth of the pool-centre headroom and removes a tenth of the patient-count gap; on BRACS neither quantity is distinguishable from zero. Recovery is the gain divided by the headroom at the same patient count; the share is the fraction of the real patient-count gap that the shrunk arms remove. Gaps are in percentage points. The BRACS ratios divide noisy differences estimated on 24 test patients per split and should be read only as bounds.

Cohort whitening remains the stronger remedy on TCGA-UT. It gains 2.26, 1.85, and 1.27 points at five, ten, and twenty patients, so At trails it by 0.97 (−1.21 to −0.74), 1.15, and 0.86 points. On BRACS, where cohort whitening gains nothing measurable, At and RWC are indistinguishable (0.05 points at five patients, −0.68 to 0.66). On TCGA-UT, directional shrinkage thus closes about half of the distance between radial shrinkage and cohort whitening at five patients, and a quarter to a third at ten and twenty, without overtaking whitening anywhere.

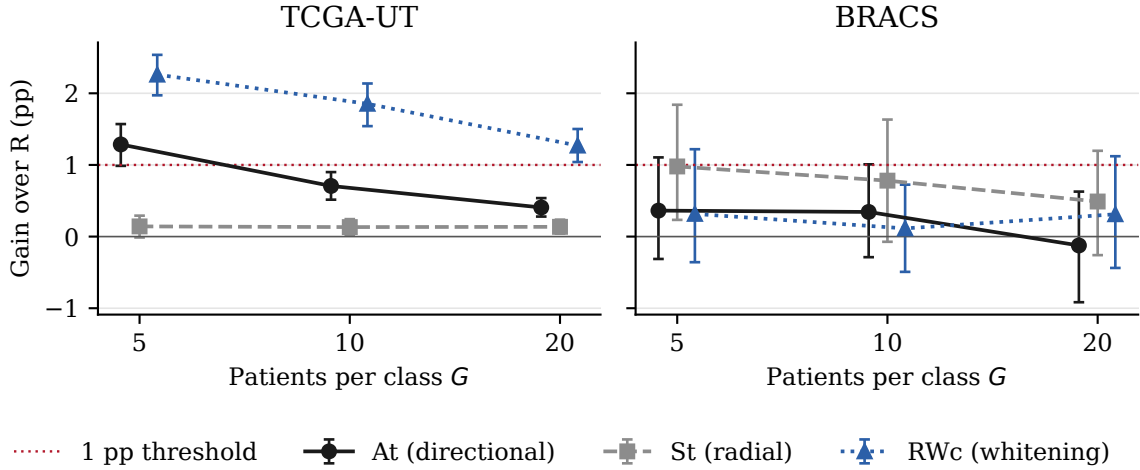


Figure 1: Directional shrinkage sits between radial shrinkage and cohort whitening on TCGA-UT and is indistinguishable from zero on BRACS. Paired gains over the real cohort in percentage points; error bars show the paired 95% bootstrap intervals. Only the TCGA-UT five-patient At point clears the one-point threshold.

5.3 Selected strengths and basis rank

Validation selection behaves differently on the two datasets (Table 4). On TCGA-UT it never selects $\alpha = 0$ and moves to stronger shrinkage as the patient count grows: the closed-form

strength or twice it at five patients, $\alpha = 2$ in 27 of 30 fits at ten, and $\alpha = 2$ or the grid maximum $\alpha = 4$ at twenty, where 12 of 30 fits sit on the upper boundary. On BRACS, $\alpha = 0$ is chosen in 5 to 9 of 30 fits per patient count and the halved strength is the most frequent or joint most frequent choice, so the validation data often prefer weak shrinkage or none.

Dataset	G	Rank	$\alpha = 0$	0.5	1	2	4
TCGA-UT	5	120	0	0	14	12	4
	10	270	0	1	1	27	1
	20	570	0	0	0	18	12
BRACS	5	28	9	12	5	2	2
	10	63	8	10	7	5	0
	20	133	5	10	10	5	0

Table 4: TCGA-UT validation data favour strong directional shrinkage; BRACS validation data favour weak shrinkage or none. Counts are tuned classifiers out of 30 per dataset and patient count (three splits, ten draws). The rank is that of the cohort’s pooled between-patient covariance, which equals its maximum $C(G - 1)$ in every draw.

The covariance rank is always full, so the eigenbasis spans 120 to 570 of the 2,560 feature dimensions on TCGA-UT and 28 to 133 on BRACS. On BRACS at five patients, a third of the eigen-directions have no estimated separation beyond noise ($s_j = 0$) and are shrunk completely at $\alpha = 1$; on TCGA-UT fewer than one in a hundred are.

5.4 What the directional correction changes

A diagnostic on the exact fitted cohorts, computed after all fits were locked, compares each estimator’s centres with the pool centres (Table 5). Following Queisler (2026b), it splits the squared centre error into a part shared by all classes, a class-specific part inside the $(C - 1)$ -dimensional discriminant subspace spanned by the centred pool centres, and a class-specific part outside it. Directional shrinkage preserves the grand mean, so it cannot change the shared part.

Dataset	G	$\bar{a}$	Off-span	Δ squared error			A: Δ by part	
				A	At	S	Disc.	Off
TCGA-UT	5	0.41	0.26	−4%	+41%	−19%	+6%	−10%
	10	0.19	0.15	−6%	+33%	−10%	+5%	−10%
	20	0.10	0.11	−3%	+62%	−3%	+5%	−8%
BRACS	5	0.78	0.39	−4%	−6%	−26%	+10%	−14%
	10	0.58	0.34	+4%	+5%	−8%	+19%	−16%
	20	0.45	0.35	+15%	+13%	+8%	+25%	−10%

Table 5: Directional shrinkage barely reduces centre error, and the tuned strength on TCGA-UT increases it. $\bar{a} = \sum_j n_j a_j / \sum_j n_j$ is the noise-weighted mean weight at $\alpha = 1$. Off-span is the fraction of the squared correction $\sum_c \|\tilde{\mu}_c - \hat{\mu}_c\|^2$ at $\alpha = 1$ that lies outside the span of the cohort’s centred class centres. Δ squared error is the relative change of $\sum_c \|\cdot - \mu_c^{\text{pool}}\|^2$ against the raw cohort centres, for the closed-form arm (A), the tuned arm at its selected strength (At), and closed-form radial shrinkage (S). The last two columns split the change of arm A into its discriminant and off-discriminant parts, as percentages of the raw squared error. Values are means over three splits and ten draws.

In one respect the estimator does what its construction promises. A quarter of its correction

at five TCGA-UT patients, and 39 percent on BRACS, lies outside the span of the cohort’s own class centres, the region that radial shrinkage cannot reach at all. Measured against the pool’s geometry, it removes off-discriminant error worth 10 and 14 percent of the raw squared error at five patients. But it adds discriminant error of 6 and 10 percent in exchange, so the net reduction in squared centre error is only 4 percent on both datasets, a fifth of what radial shrinkage achieves. On BRACS the added discriminant error grows with the patient count, and the net change turns positive at ten patients.

The accuracy gains do not follow the centre error. On TCGA-UT at five patients, radial shrinkage removes 19 percent of the squared centre error and loses 0.30 points, closed-form directional shrinkage removes 4 percent and gains 1.03 points, and the tuned arm increases squared centre error by 41 percent and gains 1.29 points. At every patient count, the tuned arm’s added error lies almost entirely inside the discriminant subspace, and it still gains the most. Across the 30 draws per patient count, the correlation between the gain of arm A and its reduction in squared centre error lies between 0.01 and 0.19 on TCGA-UT. Whatever the tuned arm improves, it is not the distance of the class centres from the pool centres.

One mechanism consistent with these observations is a first-moment analogue of whitening. Shrinking the class centres along the directions in which the cohort’s patients vary most contracts class separation along exactly those directions. The classifier then has less reason to rely on them, which is what whitening along the cohort’s between-patient basis achieves by rescaling the features (Queisler, 2026d). On this reading, validation prefers strong shrinkage because it discards unreliable patient directions, not because it moves the centres closer to the truth. The reading also matches where the remedy works: 1.29 points on TCGA-UT, where cohort whitening gains 2.26, and nothing on BRACS, where cohort whitening also gains nothing. The fits do not test this reading directly; it is an interpretation of the diagnostic, not a measured decomposition.

6 Discussion

Directional shrinkage is the first cohort-only centre correction in this series to clear the practical threshold on TCGA-UT, and it meets the prespecified criterion: 1.29 points at five patients, with an interval from 0.99 to 1.57. It recovers 12 percent of the pool-centre headroom, where radial shrinkage recovered none, and it narrows the five-to-twenty patient gap by 9 percent. On BRACS it recovers nothing measurable, and radial shrinkage remains the better estimator there. The result is therefore dataset-specific: success on TCGA-UT, a subthreshold and inconclusive effect on BRACS, and no support for a claim across cohorts.

The diagnostic changes what the TCGA-UT success means. The estimator was motivated as a better estimate of the class centres, yet its closed-form version reduces squared centre error only slightly and its tuned version increases it substantially. The gain is thus not evidence that a cohort can estimate its own class centres better from its own patients; on that question, the geometry here supports the conclusion of Queisler (2026c). What the cohort can do is decide which directions of its class-centre configuration not to trust. That is a second-moment operation expressed through the first moment, and it behaves like the second-moment remedy: helpful on TCGA-UT, ineffective on BRACS, and weaker than cohort whitening itself.

Two practical conclusions follow. First, for a five-patient TCGA-UT cohort, cohort whitening remains the better cohort-only remedy, by about one point at every patient count. Whether the two remedies add up is not tested here, and the shared mechanism suggested above makes a large combined gain unlikely. Second, the centre-error headroom of eight to eleven points remains almost entirely unrecovered. Two cohort-only centre estimators have now been tested against it, and the better one recovers 12 percent on one dataset while leaving centre error nearly unchanged. This strengthens the conclusion that the missing centre information must

come from outside the labelled cohort, for example from related classes, unlabelled slides, or a population covariance used as a prior.

7 Limitations

The estimator pools the between-patient covariance across classes, which assumes that all classes share the same patient-level noise. The eigenbasis and its eigenvalues are estimated from $C(G-1)$ patient deviations in 2,560 dimensions, only 28 for BRACS at five patients, and the separation estimate s_j uses $C-1$ degrees of freedom per direction, six on BRACS. Both estimates are noisy, and directions outside the estimated span, most of the feature space, are never shrunk. Weights that borrow strength across directions, or a covariance estimated outside the cohort, could behave differently.

The strength grid stops at $\alpha = 4$, which 12 of 30 tuned fits at twenty TCGA-UT patients selected, so the tuned arm may be limited by the grid at that patient count. The grid was not extended after test results were inspected. The primary contrast is less affected: at five patients, 4 of 30 fits chose the boundary. The TCGA-UT lower interval bound of 0.99 points does not exclude a gain slightly below the practical threshold.

The geometric diagnostic uses the pool centres as the reference for the true class centres. These centres are themselves estimated from the eligible training patients of each split, so they carry sampling error and share patients with the cohorts. The diagnostic is descriptive and has no intervals, and its mechanistic reading in Section 5.4 is an interpretation rather than a tested decomposition. All arms reuse the three evaluation splits of the source studies, so they share test patients with every earlier comparison in this series. The results are conditional on frozen Virchow2 features, multinomial logistic regression, 160 patches per class, and the two cohorts studied; transfer to other cohorts, feature extractors, or class counts is untested.

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